

Conditional Opt-Out Guarantees for Correlated Data Acquisition: A Corrected Partial-Progress Note on the Privacy-Paradox Open Problem

Prepared as a corrected verification and replacement of the draft note

June 23, 2026

Abstract

This note corrects and replaces an earlier proposed solution note about correlation-aware opt-out and externality pricing. The result below is *partial progress, not a full solution* of the open problem posed by Liao, Su, Ziani, Wierman, and Huang. The unresolved part of that open problem is to design and audit an intervention that truly prevents, bounds, or prices inference about nonparticipants when data are correlated. What is proved here is the conditional mechanism-design consequence of such an intervention.

In the Liao–Su–Ziani–Wierman–Huang model, replace the nonparticipant leakage term g_i by a residual leakage term ℓ_i . For any fixed desired participation cutoff and allocation rule, the truthful payment formula is the same as in the original paper with $g_i(\hat{c}_i)$ replaced by $\ell_i(\hat{c}_i)$. Hence lowering residual leakage from m_i to ℓ_i raises the per-selected payment by exactly $(m_i(\hat{c}_i) - \ell_i(\hat{c}_i))/A_i(c)$ and raises expected selected payments by exactly the participation mass times $m_i(\hat{c}_i) - \ell_i(\hat{c}_i)$. The fixed-participation optimization in the original paper is therefore recovered by replacing the boundary term $g_i(\hat{c}_i)$ by $\ell_i(\hat{c}_i)$ in the budget accounting; the virtual-cost term is unchanged. If $0 \leq \ell_i(c) \leq \varepsilon c$, then the remaining payment and budget distortions are $O(\varepsilon)$. A separate theorem bounds the extra participation caused by a small residual leakage cap when the exact-opt-out payment menu is held fixed. A simple perfect-correlation barrier shows that exact zero leakage cannot be imposed model-independently unless the use of correlated participant data becomes uninformative. Finally, an exact Pigouvian tariff aligns private and social objectives only under the strong assumption that residual externalities are observable or certifiably estimable.

1 Status and relation to the open problem

The paper of Liao, Su, Ziani, Wierman, and Huang studies a data-acquisition market in which one person’s data can reveal information about correlated nonparticipants. This worsens the outside option of nonparticipants and can encourage oversharing. The paper then designs truthful mechanisms for a bias–variance objective under a budget constraint. Its concluding open problem asks what approaches can mitigate information leakage and the privacy paradox in data marketplaces.

The present note does not solve that full problem. It isolates a conditional statement that any successful mitigation rule would imply. The rule is modeled by replacing the unregulated nonparticipant leakage function g_i by a residual leakage function ℓ_i . Exact protected opt-out means $\ell_i \equiv 0$; approximate protected opt-out means $0 \leq \ell_i(c) \leq \varepsilon c$. Under these assumptions one can rigorously quantify how the truthful payments, budget, and cutoff incentives change. This is useful because it identifies exactly where information leakage enters the original mechanism, but it leaves open the main implementation question: how to enforce, audit, or compute such an ℓ_i in a correlated population.

What is fixed relative to the earlier draft. The earlier draft correctly identified the boundary-payment effect of removing $g_i(\hat{c}_i)$, but it overstated the result as a solution. It also conflated two different comparative statics: a fixed-cutoff calculation in which payments are adjusted, and a fixed-menu calculation in which the cutoff moves. The present note separates these statements, adds the missing cutoff-regularity condition, and accounts for the budget effect that is central in the original paper.

2 Model fragment

There are groups $i \in [I]$. Group i has cost distribution F_i with density f_i on $\mathcal{C} = [c_{\min}, c_{\max}]$. We fix a participation-rate profile θ , so all functions below are evaluated at this profile and at the relevant correlation parameters. Thus $w = w(\bar{\theta})$ is fixed, and for group i

$$h_i(c) = h_i(c, \theta; \alpha_i) = b_i c, \quad 0 \leq b_i \leq 1.$$

A direct mechanism offers a selection probability $A_i(r)$ and a payment $P_i(r)$ to an agent who joins and reports cost r . If an agent of true cost c joins and reports r , her interim utility is

$$\bar{u}_i(r | c) = A_i(r)(P_i(r) - c + h_i(c)) - h_i(c) + w. \quad (1)$$

If she does not join, her utility is $-\ell_i(c)$, where ℓ_i is the residual leakage after regulation or intervention. The unregulated model has $\ell_i = g_i$.

For a desired cutoff $\hat{c}_i$, types $c \leq \hat{c}_i$ should participate, and types $c > \hat{c}_i$ should not. Write

$$\theta_i = F_i(\hat{c}_i),$$

where, for simplicity of notation, $F_i(c_{\min}) = 0$. The corresponding formulas with $F_i(\hat{c}_i) - F_i(c_{\min})$ are identical.

Assumption 1 (Cutoff regularity). *For each group i under consideration:*

- (i) $A_i : \mathcal{C} \rightarrow (0, 1]$ is nonincreasing and measurable;
- (ii) $\ell_i : \mathcal{C} \rightarrow \mathbb{R}_+$ is continuous;
- (iii) $c \mapsto h_i(c) - \ell_i(c)$ is nondecreasing.

The last condition is the residual-leakage analogue of the monotonicity condition used in the original paper for $h_i - g_i$. Exact protected opt-out, $\ell_i \equiv 0$, satisfies it automatically because $h_i(c) = b_i c$ is nondecreasing. If an approximate leakage rule ℓ_i violates this condition, the payment formula below still gives the correct boundary algebra, but it need not induce a single cutoff set.

3 Truthful payments with a residual leakage outside option

Fix a group i and suppress the group subscript where no confusion can arise. Set

$$Q(c) = (1 - b)A(c) + b. \quad (2)$$

The quantity Q is the effective allocation from the agent's cost perspective: using $h(c) = bc$, Eq. (1) can be rewritten as

$$\bar{u}(r | c) = A(r)P(r) - Q(r)c + w. \quad (3)$$

Lemma 1 (Envelope characterization). *Let A be nonincreasing and positive. A payment rule P is truthful for participants if and only if the truthful utility $U(c) := \bar{u}(c \mid c)$ is absolutely continuous and satisfies*

$$U'(c) = -Q(c) \quad \text{for a.e. } c \in \mathcal{C}. \quad (4)$$

Equivalently, for some constant K ,

$$U(c) = K + \int_c^{c_{\max}} Q(z) dz. \quad (5)$$

Proof. Equation (3) is a one-dimensional procurement utility with allocation Q . Since A is nonincreasing, Q is nonincreasing. If truthful reporting is optimal, the usual envelope theorem gives (4). Conversely, suppose (4) holds. The utility from reporting r when the true cost is c is

$$U(r) + Q(r)(r - c).$$

If $r > c$, then

$$U(c) - U(r) - Q(r)(r - c) = \int_c^r (Q(z) - Q(r)) dz \geq 0.$$

If $r < c$, then

$$U(c) - U(r) - Q(r)(r - c) = \int_r^c (Q(r) - Q(z)) dz \geq 0.$$

Thus reporting truthfully maximizes utility. The integral representation (5) is the integrated form of (4). $\square$

Theorem 1 (Cutoff-normalized payment formula). *Under [Assumption 1](#), the cutoff-normalized truthful payment rule that induces cutoff $\hat{c}$ is*

$$P^\ell(c) = c - h(c) + \frac{(1 - b) \int_c^{c_{\max}} A(z) dz + \tau^\ell}{A(c)}, \quad c \in \mathcal{C}, \quad (6)$$

where

$$\tau^\ell = h(\hat{c}) - \ell(\hat{c}) - (1 - b) \int_{\hat{c}}^{c_{\max}} A(z) dz - w. \quad (7)$$

The truthful joining utility is

$$U^\ell(c) = (1 - b) \int_c^{c_{\max}} A(z) dz + \tau^\ell - h(c) + w. \quad (8)$$

Moreover,

$$U^\ell(c) + \ell(c) = (1 - b) \int_c^{\hat{c}} A(z) dz + (h(\hat{c}) - \ell(\hat{c})) - (h(c) - \ell(c)). \quad (9)$$

Hence every $c < \hat{c}$ participates and every $c > \hat{c}$ does not participate, with indifference at $c = \hat{c}$; weak inequalities hold if the monotonicity in [Assumption 1](#) is weak.

Proof. By [Lemma 1](#), truthful utility has the form

$$U(c) = K + \int_c^{c_{\max}} ((1 - b)A(z) + b) dz.$$

The cutoff type must be indifferent between joining and not joining, so $U(\hat{c}) = -\ell(\hat{c})$. Solving for K gives

$$U(c) = -\ell(\hat{c}) + \int_c^{\hat{c}} ((1 - b)A(z) + b) dz,$$

which is equivalent to (8) with τ^ℓ as in (7). Solving the identity

$$U(c) = A(c)(P(c) - c + h(c)) - h(c) + w$$

for $P(c)$ gives (6). Adding $\ell(c)$ to $U^\ell(c)$ and using $h(c) = bc$ gives (9). If $c < \hat{c}$, then the integral term in (9) is nonnegative, and cutoff regularity gives

$$h(\hat{c}) - \ell(\hat{c}) \geq h(c) - \ell(c).$$

Thus $U^\ell(c) + \ell(c) \geq 0$. If $c > \hat{c}$, both inequalities reverse, so $U^\ell(c) + \ell(c) \leq 0$. This is exactly the desired participation cutoff. $\square$

Remark 1 (Minimality). *For a fixed allocation rule A and cutoff $\hat{c}$, the boundary condition $U(\hat{c}) = -\ell(\hat{c})$ pins down the lowest truthful utility compatible with participation of all types below the cutoff. Raising the integration constant raises payments and leaves truthfulness intact; lowering it makes the cutoff type reject the mechanism. This is the sense in which Theorem 1 is cutoff-normalized and minimal.*

4 Exact comparison with the unregulated model

Let m and ℓ be two residual leakage functions satisfying Assumption 1; m may be the unregulated leakage g , and ℓ may be a regulated residual leakage. Use the same allocation rule A and cutoff $\hat{c}$ in both mechanisms.

Proposition 1 (Boundary leakage is the entire payment difference). *For every cost c with $A(c) > 0$,*

$$P^\ell(c) - P^m(c) = \frac{m(\hat{c}) - \ell(\hat{c})}{A(c)}. \quad (10)$$

In particular, exact protected opt-out $\ell \equiv 0$ removes the unregulated leakage subsidy:

$$P^0(c) - P^g(c) = \frac{g(\hat{c})}{A(c)}. \quad (11)$$

Proof. The payment formula (6) is identical for m and ℓ except for the boundary term $-m(\hat{c})$ or $-\ell(\hat{c})$ in τ . Subtraction gives (10) immediately. $\square$

The per-selected payment gap in (10) is divided by $A(c)$ because $P(c)$ is paid only conditional on selection. The expected selected-payment gap is simpler.

Proposition 2 (Budget accounting). *For group i , define the expected selected payment per group member by*

$$\mathcal{B}_i^\ell(A_i, \hat{c}_i) = \int_{c_{\min}}^{\hat{c}_i} A_i(c) P_i^\ell(c) f_i(c) dc. \quad (12)$$

Then

$$\mathcal{B}_i^\ell(A_i, \hat{c}_i) - \mathcal{B}_i^m(A_i, \hat{c}_i) = F_i(\hat{c}_i)(m_i(\hat{c}_i) - \ell_i(\hat{c}_i)). \quad (13)$$

Consequently, for group weights q_i and population size s , the total expected selected-payment increase from m to ℓ is

$$s \sum_i q_i F_i(\hat{c}_i)(m_i(\hat{c}_i) - \ell_i(\hat{c}_i)). \quad (14)$$

Proof. Multiply (10) by $A_i(c)$ and integrate over $c \leq \hat{c}_i$:

$$\int_{c_{\min}}^{\hat{c}_i} A_i(c)(P_i^\ell(c) - P_i^m(c))f_i(c) dc = (m_i(\hat{c}_i) - \ell_i(\hat{c}_i)) \int_{c_{\min}}^{\hat{c}_i} f_i(c) dc.$$

This proves (13); multiplying by sq_i and summing over groups proves (14). $\square$

This budget effect is not a cosmetic issue. If the platform's budget is fixed, then eliminating nonparticipant leakage while holding the same participation cutoffs and allocation rule requires additional selected-payment budget. Thus exact opt-out protection may force a lower participation rate, lower selection probabilities, or a larger budget. The original draft omitted this point.

5 Connection with the original virtual-cost optimization

The preceding budget identity can also be written in the same form as the fixed-participation optimization in the original paper. For group i , define

$$\phi_i(c) = c - h_i(c) + (1 - b_i) \frac{F_i(c)}{f_i(c)}. \quad (15)$$

This is the same virtual cost as in the original model; it depends on the participant privacy cost h_i , not on the outside-option leakage term.

Proposition 3 (Residual leakage only changes the boundary budget term). *For any residual leakage ℓ_i satisfying Assumption 1, the total expected selected payment per capita induced by Theorem 1 is*

$$\frac{1}{s} \mathcal{B}^\ell(A, \hat{c}) = \sum_i q_i \int_{c_{\min}}^{\hat{c}_i} A_i(c) \phi_i(c) f_i(c) dc + L^\ell(\hat{c}), \quad (16)$$

where

$$L^\ell(\hat{c}) = \sum_i q_i F_i(\hat{c}_i) (h_i(\hat{c}_i) - \ell_i(\hat{c}_i) - w). \quad (17)$$

Therefore, for every fixed participation profile, the fixed- θ allocation optimization of the original paper is obtained by replacing the original boundary term $g_i(\hat{c}_i)$ by $\ell_i(\hat{c}_i)$ in L^ℓ . The virtual costs ϕ_i and the monotonicity/truthfulness analysis are unchanged.

Proof. Using (6),

$$A_i(c)P_i^\ell(c) = A_i(c)(c - h_i(c)) + (1 - b_i) \int_c^{c_{\max}} A_i(z) dz + \tau_i^\ell.$$

Integrating over $c \in [c_{\min}, \hat{c}_i]$ gives

$$\begin{aligned} \int_{c_{\min}}^{\hat{c}_i} A_i(c)P_i^\ell(c)f_i(c) dc &= \int_{c_{\min}}^{\hat{c}_i} A_i(c)(c - h_i(c))f_i(c) dc \\ &\quad + (1 - b_i) \int_{c_{\min}}^{\hat{c}_i} \left(\int_c^{c_{\max}} A_i(z) dz \right) f_i(c) dc + F_i(\hat{c}_i)\tau_i^\ell. \end{aligned}$$

By Fubini's theorem,

$$\int_{c_{\min}}^{\hat{c}_i} \left(\int_c^{c_{\max}} A_i(z) dz \right) f_i(c) dc = \int_{c_{\min}}^{\hat{c}_i} A_i(z)F_i(z) dz + F_i(\hat{c}_i) \int_{\hat{c}_i}^{c_{\max}} A_i(z) dz.$$

The last term cancels with the corresponding term in $F_i(\hat{c}_i)\tau_i^\ell$. After collecting terms one obtains

$$\int_{c_{\min}}^{\hat{c}_i} A_i(c) \left(c - h_i(c) + (1 - b_i) \frac{F_i(c)}{f_i(c)} \right) f_i(c) dc + F_i(\hat{c}_i)(h_i(\hat{c}_i) - \ell_i(\hat{c}_i) - w).$$

Multiplying by q_i , summing over i , and multiplying by s gives (16). $\square$

Corollary 1 (Effective-budget loss from protected opt-out). *Let g_i be the unregulated leakage and let ℓ_i be a regulated residual leakage. For the same fixed cutoffs and allocation rule,*

$$L^\ell(\hat{c}) - L^g(\hat{c}) = \sum_i q_i F_i(\hat{c}_i)(g_i(\hat{c}_i) - \ell_i(\hat{c}_i)). \quad (18)$$

In the exact protected-opt-out case $\ell_i \equiv 0$, the selected-payment budget needed to maintain the same fixed participation profile increases by

$$s \sum_i q_i F_i(\hat{c}_i) g_i(\hat{c}_i). \quad (19)$$

Thus the conditional mitigation does not create free participation. It removes the leakage subsidy and forces the platform either to pay for the participation it induces or to reduce acquisition.

6 Approximate protected opt-out

There are two distinct approximate statements. The first holds when the target cutoff is fixed and payments are recomputed. The second holds when the exact opt-out menu is fixed and residual leakage changes participation behavior.

6.1 Fixed cutoff, recomputed payments

Proposition 4 (Payment and budget distortion under a leakage cap). *Assume $0 \leq \ell_i(c) \leq \varepsilon c$ for all c and $A_i(c) \geq a_i > 0$ on $[c_{\min}, \hat{c}_i]$. For the fixed cutoff and allocation rule,*

$$0 \leq P_i^0(c) - P_i^\ell(c) = \frac{\ell_i(\hat{c}_i)}{A_i(c)} \leq \frac{\varepsilon c_{\max}}{a_i}, \quad c \leq \hat{c}_i. \quad (20)$$

The expected selected-payment difference per group member satisfies

$$0 \leq \mathcal{B}_i^0(A_i, \hat{c}_i) - \mathcal{B}_i^\ell(A_i, \hat{c}_i) = F_i(\hat{c}_i)\ell_i(\hat{c}_i) \leq F_i(\hat{c}_i)\varepsilon c_{\max}. \quad (21)$$

Proof. The payment identity is Proposition 1 with $m = \ell$ and residual leakage 0 as the comparison benchmark. The upper bound follows from $\ell_i(\hat{c}_i) \leq \varepsilon \hat{c}_i \leq \varepsilon c_{\max}$ and $A_i(c) \geq a_i$. The budget identity is Proposition 2 with the same substitution. $\square$

6.2 Fixed exact-opt-out menu, moving cutoff

Now consider a different experiment. Suppose the platform posts the payment menu designed for exact protected opt-out, but residual leakage ℓ remains. Because nonparticipation is now worse than intended, some extra types may join. This is not the same as Proposition 4, where payments were recomputed to hold the cutoff fixed.

Proposition 5 (Cutoff expansion from residual leakage). *Fix a group and an exact-opt-out truthful menu with truthful utility U^0 and cutoff c^0 , so $U^0(c^0) = 0$ and $U^0(c) < 0$ for $c > c^0$. Suppose*

$$Q(c) = (1 - b)A(c) + b \geq \rho > 0 \quad \text{for all } c \in [c^0, c_{\max}]. \quad (22)$$

If residual leakage satisfies $0 \leq \ell(c) \leq \delta$ for all c , then no type $c > c^0 + \delta/\rho$ participates under the same menu. In particular, if the new participation set is an interval $[c_{\min}, c^\ell]$, then

$$0 \leq c^\ell - c^0 \leq \frac{\delta}{\rho}. \quad (23)$$

With the cap $\ell(c) \leq \varepsilon c$, one may take $\delta = \varepsilon c_{\max}$.

Proof. By the envelope identity, $(U^0)'(c) = -Q(c) \leq -\rho$ almost everywhere. Hence for $c \geq c^0$,

$$U^0(c) \leq -\rho(c - c^0).$$

A type c participates under residual leakage only if $U^0(c) + \ell(c) \geq 0$. If $c > c^0 + \delta/\rho$, then

$$U^0(c) + \ell(c) < -\delta + \delta = 0,$$

so such a type cannot participate. This proves the claimed bound. $\square$

7 Why exact zero leakage cannot be unconditional

The formulas above are conditional on the existence of a residual leakage function ℓ_i satisfying the desired bound. The following elementary example explains why the exact requirement $\ell_i \equiv 0$ cannot be obtained from an arbitrary useful statistical release when correlations are unrestricted.

Proposition 6 (Perfect-correlation barrier). *Let X_o be the secret of an opt-out individual and X_p the datum of a participant. Suppose $X_o = X_p$ almost surely and X_p is not almost surely constant. Let Y be any randomized output, model score, or downstream action produced from X_p by an arbitrary Markov kernel. Then*

$$I(Y; X_o) = I(Y; X_p), \quad (24)$$

where $I(\cdot; \cdot)$ denotes mutual information. Consequently, an exact opt-out guarantee requiring $I(Y; X_o) = 0$ forces $I(Y; X_p) = 0$.

Proof. Because $X_o = X_p$ almost surely, the joint distributions of (Y, X_o) and (Y, X_p) are identical. Mutual information depends only on the joint distribution, so (24) follows. If exact opt-out protection requires $I(Y; X_o) = 0$, then the same equality implies $I(Y; X_p) = 0$. $\square$

Thus a model-independent exact opt-out guarantee is too strong in the presence of arbitrary correlation: in the perfectly correlated case, learning anything in mutual-information terms about the participant is learning the same amount about the opt-out. This does not rule out useful regulation. It shows that a full solution must add structure, such as a restricted correlation class, an explicit accuracy loss, a weaker leakage metric, institutional purpose limitations, or pricing and compensation for residual leakage.

8 Externality pricing: valid only as a conditional implementation result

The opt-out calculation above concerns the outside option of potential participants. A separate Pigouvian tariff can internalize residual harms to people whose data or inference rights have not been purchased. This statement is mathematically correct but strong: it assumes the regulator can observe or certifiably estimate the externalities being priced.

Theorem 2 (Conditional Pigouvian alignment). *Let x denote any feasible data-acquisition and data-use policy. Let $V(x)$ be platform value, let $d_j(x)$ be the direct costs already compensated through licenses or participant payments, and let $e_j(x)$ be the residual unpriced correlation externality imposed on person j . If the platform is charged*

$$T(x) = \sum_j e_j(x), \quad (25)$$

then its regulated payoff, up to transfers among agents, is

$$V(x) - \sum_j d_j(x) - T(x) = V(x) - \sum_j d_j(x) - \sum_j e_j(x), \quad (26)$$

which is the corresponding utilitarian welfare objective. Hence every profit-maximizing platform policy maximizes this welfare objective over the same feasible set.

Proof. Substitute (25) into the platform payoff. The resulting expression is exactly the welfare objective in (26), so private and social rankings of feasible policies coincide. $\square$

In the continuum group notation, a natural residual opt-out tariff corresponding to ℓ_i is

$$\mathcal{T}^\ell(\hat{c}) = s \sum_i q_i \int_{\hat{c}_i}^{c_{\max}} \ell_i(c) f_i(c) dc, \quad (27)$$

possibly plus terms for downstream uses not covered by participant licenses. The theorem says that if these residual harms are correctly measured and charged, the platform internalizes them. It does not say that such measurement is easy, or even that a proposed audit metric equals the true utility cost. That remains part of the open problem.

9 What has and has not been obtained

The results above give the following rigorous partial progress.

- (1) The exact mathematical role of nonparticipant leakage in the original payment formula is isolated: only the boundary term $g_i(\hat{c}_i)$ changes when the residual outside-option leakage changes.
- (2) The payment effect is exact. Lowering residual leakage from m_i to ℓ_i raises the per-selected payment by $(m_i(\hat{c}_i) - \ell_i(\hat{c}_i))/A_i(c)$.
- (3) The budget effect is exact. The expected selected-payment budget needed to maintain the same cutoff and allocation increases by participation mass times the removed boundary leakage.

- (4) For a fixed participation profile, the original virtual-cost allocation problem is modified only by replacing the boundary term $g_i(\hat{c}_i)$ by $\ell_i(\hat{c}_i)$ in the budget accounting. The virtual cost itself is not changed by the opt-out outside option.
- (5) If a regulator can enforce $0 \leq \ell_i(c) \leq \varepsilon c$, then the residual payment distortion and the fixed-menu cutoff expansion are bounded linearly in ε .
- (6) A perfect-correlation example shows why exact zero leakage cannot be expected as a model-independent guarantee while preserving nontrivial use of correlated participant data.

However, this is not a full solution of the Liao–Su–Ziani–Wierman–Huang open problem. The following issues remain unresolved here.

- (a) No technical privacy mechanism is constructed that guarantees $\ell_i \equiv 0$ or $\ell_i(c) \leq \varepsilon c$ for correlated nonparticipants while preserving a desired statistical task.
- (b) No audit procedure is proved to estimate ℓ_i or e_j accurately in equilibrium. A posterior-risk, mutual-information, or Pufferfish-style metric may be useful, but this note does not prove that any such metric equals agents’ utility costs.
- (c) The full endogenous optimization over participation profiles under the modified budget is not solved. The fixed-profile optimization is reduced to the original formulas with a changed boundary term, but the best regulated participation profile may change.
- (d) Strategic behavior by multiple platforms, data brokers, or downstream users is not modeled.

Thus the corrected status is: *conditional partial progress*. The note shows what follows from a successful leakage-limiting or leakage-pricing intervention; it does not itself supply the missing intervention.

A Verification against the earlier draft

The following points summarize the mathematical verification.

- (i) The earlier payment identity $P_i^0(c) - P_i^g(c) = g_i(\hat{c}_i)/A_i(c)$ is correct for a fixed allocation rule and fixed cutoff, assuming $A_i(c) > 0$.
- (ii) The identity must be stated conditionally. It does not construct a privacy mechanism that makes g_i vanish; it only describes the consequence if residual leakage is reduced.
- (iii) A cutoff theorem needs the regularity condition that $h_i(c) - \ell_i(c)$ be nondecreasing. Without such a condition, setting the cutoff type indifferent does not guarantee that all lower costs join and all higher costs do not.
- (iv) The fixed-cutoff payment comparison and the fixed-menu cutoff-shift comparison are different experiments. They are separated in [Propositions 4](#) and [5](#).
- (v) The original draft omitted the expected-budget increase. The exact correction is [Propositions 2](#) and [3](#): removing leakage removes a subsidy, so the platform must either spend more selected-payment budget or acquire less data.
- (vi) The Pigouvian argument is valid only under exact externality measurement or a certified estimator of that externality. It is an alignment lemma, not by itself a solution to the measurement and enforcement problem.

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